



Haonan Dong

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EDUCATION AND TRAINING

2026/09/24 – CURRENT Munich, Germany

MASTER OF SCIENCE, GEODESY AND GEOINFORMATION Technical University of Munich

2020/08/31 – 2023/05/31 Wuhan, China

MASTER OF ENGINEERING, PATTERN RECOGNITION AND INTELLIGENT SYSTEMS Wuhan University

2019/08/31 – 2019/11/30 Santa Barbara, CA, United States

UNDERGRADUATE EXCHANGE PROGRAM, COMPUTER SCIENCE University of California, Santa Barbara (UCSB)

2016/08/31 – 2020/05/31 Wuhan, China

BACHELOR OF ENGINEERING, GEODESY AND GEOMATICS ENGINEERING Wuhan University

WORK EXPERIENCE

MID-LEVEL TECHNICAL ENGINEER – LABORATORY OF ARTIFICIAL INTELLIGENCE AND DIGITAL ECONOMY (SHENZHEN) – 2024/09/30 – 2026/09/12 – SHENZHEN, CHINA

- Developed a UAV photogrammetric path planning system supporting nadir, oblique, and optimized viewpoints for high-precision aerial data acquisition.
- Designed and implemented multi-sensor fusion algorithms by integrating visual SLAM and GNSS, enabling accurate real-time orthophoto rectification.
- Developed a large-scale 3D Gaussian Splatting reconstruction framework, enabling high-precision and distributed training for urban-scale scene modeling.

ALGORITHM ENGINEER – SHENZHEN XGRIDS INNOVATION TECHNOLOGY CO., LTD. – 2023/06/30 – 2024/08/31 – SHENZHEN, CHINA

- Developed advanced calibration modules for multi-sensor systems, including high-precision LiDAR, IMU, and camera extrinsic and intrinsic calibration.
- Designed and optimized visual modules for a handheld SLAM system, improving robustness and reconstruction quality in real-world environments.

LANGUAGE SKILLS

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **CERTIFICATIONS**

Shenzhen Human Resources and Social Security Bureau, 2024/08/31
Assistant Engineer Qualification (Geodesy & Geomatics)

● **PUBLICATIONS**

2025
[Large-Scale 3D Gaussian Splatting Based on UAV Multi-Modal Platform](#)

• **Oral Presentation:** Japan Society for Photogrammetry and Remote Sensing (JSPSRS), Fall Academic Conference, 20/11/2025, 3rd Floor, Yamagata Terrace, 1-2-3 Futaba-cho, Yamagata City, Yamagata Prefecture, 990-0828, Japan

Research Funding: ZTE Industry–Academia Collaboration Fund Project

Authors: Haonan Dong, Zhijun Wang, Qingyun Song, Yuhan Tian, Ryan Ge | **Journal Name:** Japan Society for Photogrammetry and Remote Sensing | **Volume, Issue and Pages:** Pages 25-28

2023
[Learning-Based Encoded Target Detection on Iteratively Orthorectified Images for Accurate Fisheye Calibration](#)

Research Funding: Industry–University Collaboration with DiDi Global Inc.

Authors: Haonan Dong, Jian Yao, Ye Gong, Li Li, Shaosheng Cao, Yuxuan Li | **Journal Name:** The Photogrammetric Record | **Volume, Issue and Pages:** Volume38, Issue183, Pages 297-319

2023
[View-Graph Key-Subset Extraction for Efficient and Robust Structure From Motion](#)

Research Funding: University–Enterprise Collaboration with Huawei Technologies Co., Ltd.

Authors: Ye Gong, Pengwei Zhou, Yuyue Liu, Haonan Dong, Li Li, Jian Yao | **Journal Name:** The Photogrammetric Record | **Volume, Issue and Pages:** Volume38, Issue183, Pages 252-296

2021
[Emergency Evacuation Path Planning Algorithm for Indoor Fire in Commercial Buildings](#)

Research Funding: National Undergraduate Innovation Project

Authors: Haonan Dong, Xiaotong Ye, Congpu Hao | **Journal Name:** Journal of Geomatics | **Volume, Issue and Pages:** Volume 46 Sup. 1, Pages 40-43

● **PROJECTS**

2025/07/31 – 2026/09/12
Large-Scale Scene Reconstruction Based on Multi-Modal Fusion, Module Leader, ZTE Industry–Academia Collaboration Fund Project

2023/06/30 – 2024/08/31
Multi-Sensor Fusion Algorithm Development, Module Leader, Xgrids Project

2022/02/28 – 2023/04/30
High-Accuracy RGB-D Indoor Scene Reconstruction, Project Leader, University–Enterprise Collaboration with Huawei Technologies Co., Ltd.

2021/11/30 – 2022/05/31
Calibration of Multi-Camera Systems for Autonomous Vehicles, Project Leader, Industry–University Collaboration with DiDi Global Inc.

2018/04/30 – 2019/05/31
Indoor Fire Evacuation System Based on Wireless Sensor Networks (WSN), Project Leader, National Undergraduate Innovation Project

● HONOURS AND AWARDS

2023/09/30

Best Newcomer Award – Shenzhen Xgrids Innovation Technology Co., Ltd.

2021/05/31

Outstanding Graduate Scholarship (Master) – Wuhan University

2020/05/31

Outstanding Undergraduate Graduate – Wuhan University

2020/02/29

National Undergraduate Innovation Program – Excellent Project Award (Top 1 in School of Geodesy and Geomatics) – Wuhan University

2019/05/31

Grand Prize in “South Survey Cup” Student Paper Competition” – School of Geodesy and Geomatics, Wuhan University

2019/02/28

Honorable Mention, Mathematical Contest in Modeling (MCM/ICM) – COMAP (Consortium for Mathematics and its Applications)

2018/08/31

Third Prize, National College Student Mathematical Modeling Competition (Hubei Province) – CSIAM (China Society for Industrial and Applied Mathematics)

Second-Class Undergraduate Scholarship * 3 – Wuhan University

In 2017, 2018, 2019

● PATENTS

2026/05/29 – CURRENT

Method, System, and Storage Medium for 3D Gaussian Splatting Reconstruction Directly from Colored Point Clouds [P].

Patent Information: Guangdong, China: CN202610761683.5

2024/01/31 – CURRENT

Haonan Dong. Method, Device, Equipment, and Storage Medium for Extrinsic Calibration between LiDAR and Camera [P].

Patent Information: Guangdong, China: CN202311719319.5, 2024-02-23.

Research Funding: Xgrids Project

2022/12/31 – CURRENT

Haonan Dong, Pengwei Zhou, Lei Luo. Method, Device, Equipment, and Storage Medium for Parameter Calibration of Dual Fisheye Cameras [P].

Patent Information: Guangdong, China: CN202211132179.7, 2023-01-03.

Research Funding: Industry–University Collaboration with DiDi Global Inc.

2022/10/31 – CURRENT

Haonan Dong, Pengwei Zhou, Lei Luo. Camera Calibration Method, Device, Equipment, and Storage Medium Based on a General Calibration Board [P].

Patent Information: Guangdong, China: CN202210999236.5, 2022-11-29.

Research Funding: Industry–University Collaboration with DiDi Global Inc.